Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 6)

We will go through Questions 3 & 4 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. Suppose that $\boldsymbol{z}$ and $\boldsymbol{u}$ are nonzero vectors satisfying

$$\boldsymbol{u}^\mathsf{T}\mathbf{1} = \boldsymbol{z}^\mathsf{T}\mathbf{1} = \boldsymbol{z}^\mathsf{T}\boldsymbol{u} = 0 \qquad \text{and} \qquad \boldsymbol{z}^\mathsf{T}\boldsymbol{z} = \boldsymbol{u}^\mathsf{T}\boldsymbol{u}.$$

   For a small positive constant δ, define

$$\boldsymbol{x}_1 = \boldsymbol{z} + \delta\boldsymbol{u}, \qquad \boldsymbol{x}_2 = \boldsymbol{z} - \delta\boldsymbol{u}, \qquad \boldsymbol{y} = \boldsymbol{u}.$$

   All regression models include an intercept.

   (a) Show that the sample correlation between $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ is

$$\frac{\boldsymbol{x}_1^T\boldsymbol{x}_2}{\|\boldsymbol{x}_1\|\|\boldsymbol{x}_2\|} = \frac{1-\delta^2}{1+\delta^2},$$

   (b) The coefficient of determination (R-squared) can be evaluated as

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$

   Show that each of the simple regressions of $\boldsymbol{y}$ on $\boldsymbol{x}_1$ and of $\boldsymbol{y}$ on $\boldsymbol{x}_2$ has

$$R^2 = \frac{\delta^2}{1+\delta^2}.$$

   Interpret the results from parts (a) and (b) when δ is small. In particular, comment on the correlation between the two covariates and on their individual explanatory power for $\boldsymbol{y}$.

   (c) Express $\boldsymbol{y}$ as a linear combination of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. Hence, determine the coefficients and the value of R^2 for the multiple regression of $\boldsymbol{y}$ on $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. Use your results to explain why high correlation between two covariates does not, by itself, justify removing one of them from a regression model.

2. Consider a linear regression model with design matrix $\boldsymbol{X}$ of full column rank. Suppose that observation i has residual $e_i = 0$.

   (a) Show that deleting observation i does not change the least-squares coefficient estimates, provided that the design matrix for the remaining observations has full column rank.

   (b) What effect does the deletion have on the usual unbiased estimator of the error variance?

3. The `trees` dataset in R contains measurements from 31 felled black cherry trees. The variables are:

   - `Girth`: the diameter of the tree, measured in inches;
   - `Height`: the height of the tree, measured in feet;
   - `Volume`: the volume of timber, measured in cubic feet.

   Although the first variable is called `Girth`, it is a measurement of diameter rather than circumference.

   (a) Fit the linear regression model

$$\texttt{Volume} = \beta_0 + \beta_1\texttt{Girth} + \beta_2\texttt{Height} + \varepsilon.$$

   Produce appropriate diagnostic plots and assess the assumptions of the linear regression model. Comment on any evidence of an inadequate mean structure, nonconstant error variance, or unusual observations.

(b) One possible way of allowing for curvature is to add quadratic terms. Fit the model

$$\texttt{Volume} = \beta_0 + \beta_1 \texttt{Girth} + \beta_2 \texttt{Girth}^2 + \beta_3 \texttt{Height} + \beta_4 \texttt{Height}^2 + \varepsilon.$$

Compare its diagnostic plots with those from part (a). Does the addition of quadratic terms adequately address the problems identified in part (a)? Comment briefly on whether there is a scientific reason to expect timber volume to be an additive quadratic function of diameter and height.

(c) Approximating the trunk of a tree by a cylinder suggests that its volume should be approximately proportional to its cross-sectional area multiplied by its height. This gives the relationship

$$\texttt{Volume} \approx c\, \texttt{Girth}^2 \texttt{Height},$$

where `Girth` denotes diameter. Use this relationship to motivate a linear regression model for log(`Volume`) in terms of log(`Girth`) and log(`Height`). Fit the resulting model and assess its diagnostic plots. Compare the diagnostics with those from parts (a) and (b). Compare the estimated coefficients of log(`Girth`) and log(`Height`) with the geometrically suggested values 2 and 1, respectively.

(d) Retaining log(`Girth`) and log(`Height`) as the covariates, consider the Box–Cox family of transformations of the response, defined by

$$g_\lambda(y) = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log(y), & \lambda = 0. \end{cases}$$

Use the `boxcox` function from the `MASS` package to construct a Box–Cox profile likelihood for a model of `Volume` with predictors log(`Girth`) and log(`Height`), that is

$$g_\lambda(\texttt{Volume}) = \beta_0 + \beta_1 \log(\texttt{Girth}) + \beta_2 \log(\texttt{Height}) + \varepsilon.$$

Is $\lambda = 1$, which is equivalent to using the untransformed response in a model containing an intercept, plausible? Is $\lambda = 0$, corresponding to a logarithmic transformation, plausible? Taking account of both the profile likelihood and the diagnostic plots, recommend a final model.

4. The `sat` data in the `faraway` package contain state-level educational expenditure and SAT results for the 50 US states in 1994–95. Fit the model

$$\texttt{total} = \beta_0 + \beta_1 \texttt{expend} + \beta_2 \texttt{salary} + \beta_3 \texttt{ratio} + \beta_4 \texttt{takers} + \epsilon.$$

```
library(faraway)
?sat
sat_lm <- lm(total ~ expend + salary + ratio + takers, data = sat)
```

(a) Calculate the studentised residuals. Are there any observations that might reasonably be regarded as outliers? Identify any such states and comment on the strength of the evidence.

(b) Calculate the variance inflation factors for the four covariates. Is there evidence of collinearity? If so, identify the covariates principally involved and explain what the corresponding variance inflation factors indicate about the precision of their estimated coefficients.

(c) Fit a reduced model obtained by dropping `expend`. Compare the estimated coefficient and standard error for `salary` in the two models, and recalculate the variance inflation factors. Explain the changes you observe. Does dropping `expend` materially change your conclusions about `salary`?